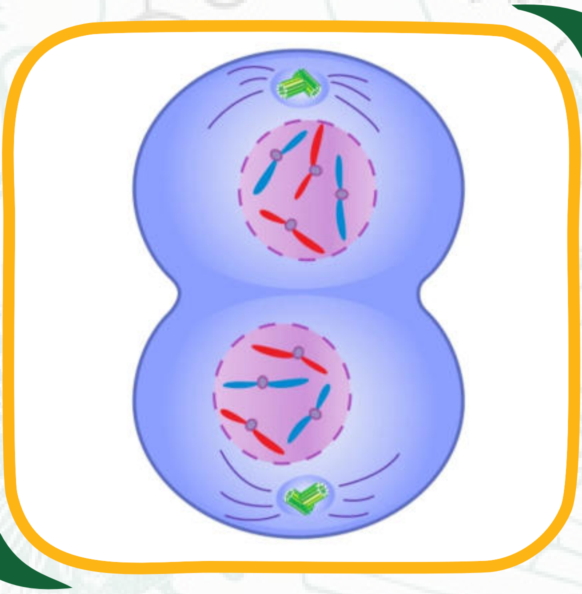
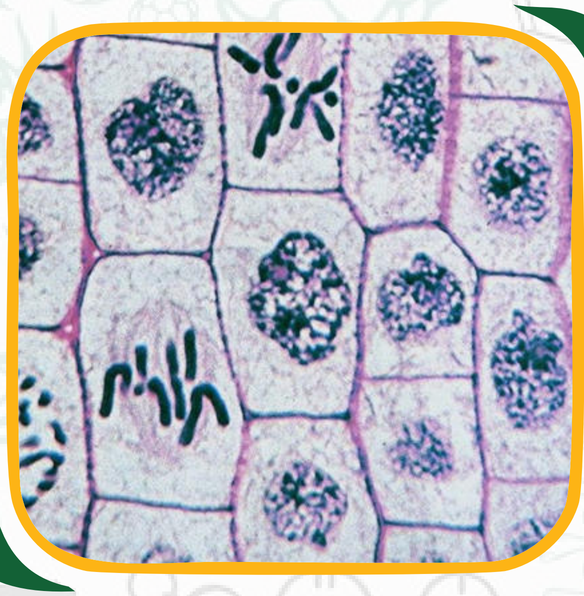
CELL CYCLE





DEFINITION

CYTOKINESIS

It is the division of cytoplasm.

KARYOKINESIS

It is the division of nucleus.

CENTROMERE

It is the region of chromosomes at which spindle fiber attached during cell division.

CHIASMATA

The homologous chromosomes which in contact at one or more points are called chiasmata.

CHROMATIDS

A chromatid is one of the two identical halves of a chromosomes that has been replicated in preparation for cell division.

SYNAPSIS

The pairing of homologous chromosomes is known as synapsis.

BIVALENT

The paired homologous chromosomes are known as bivalent.

TETRAD

Each bivalent is composed of four chromatids and therefore is known as tetrad.

CROSSING OVER

It is the process in which non-sister chromatids exchange their segments of homologous

SHORT/DETAILED ANSWER & QUESTIONS

Q. Define cell cycle. Also describe its phases.

Ans. **CELL CYCLE**

DEFINITION:

Cell cycle is the series of events from the time a cell is produced until it completes mitosis and produces new cells.

PHASES OF CELL CYCLE:

The cell cycle consists of two major phases. i.e.,

i. Interphase

ii. Mitotic phase (M-phase)

The mitotic phase is the relatively short period of cell cycle. It alternates with the much longer interphase, where the cells prepare itself for division. Typically, interphase lasts for at least 90% of the total time required for cell cycle.

Q. What do you know about M-Phase?

Ans. After the G2 phase of interphase, the cell enters the division phase i.e., M-Phase. It is characterized by mitosis in which cell divides into two daughter cells.

Q. What is difference between in cytokinesis in animal cell and plant cell.

Ans. **CYTOKINESIS IN ANIMAL CELL:**

In animal cells cytokinesis takes place by developing a constriction. Thin constriction become deep to divide cytoplasm in two equal halves and two daughter cells are formed.

CYTOKINESIS IN PLANT CELL:

In plant cell it occurs by developing cell plate. In this way the daughter cells become the exact copies of their parent cell. Resulting zygote will also have abnormal number of chromosomes.

Q. Define interphase. Explain the stages of interphase.

Ans. **INTERPHASE:**

Definition:

Interphase is the time when a cell's metabolic activity is very high, as it performs its various functions.

STAGES OF INTERPHASE:

It's divided into three phases.

i. G1 (First gap)

ii. S (Synthesis)

iii. G2 (Second gap)

G1-PHASE:

It is the period of extensive metabolic activity in which.

- Cell grows in size
- Synthesis of various enzymes
- DNA base unit are accumulated for the DNA synthesis.

S-PHASE:

In this phase, DNA replication takes place, as a result the cell duplicates its chromosomes. As a result, each chromosome consists of two sister chromatids.

G2-PHASE:

The following changes occur during this phase.

- The cell prepare protein that are essential for mitosis, mainly for the production of spindle fibers.
- Cell grows in size.
- Cell organelles are replicate in numbers.

After G2 phase of interphase, the cell enters in the division phase i.e., Mphasis.

Q. Describe various phases of mitosis with suitable diagrams.

Ans. **PHASES OF MITOSIS**

i. Prophase

ii. Metaphase

iii. Anaphase

iv. Telophase

i. PROPHASE:

i. Chromatin material condenses and appear as thread like structure called chromosomes.

ii. Centrosomes divided into two and move to the opposite side of the nucleus and give rise to spindle fibers with attach of each chromatid.

iii. Centrosomes become shorter and thicker.

iv. Each chromosome splits longitudinally into two chromatids but held together at a point called Centromere.

v. Nuclear membrane and nucleolus start disappearing and disappeared completely at the end of prophase

ii. METAPHASE:

In this phase all the chromosomes come to lie at the equatorial plane and spindle Fibers start to contract.

iii. ANAPHASE:

In this phase each chromosome divides into two separate chromatids begin to move at the opposite poles.

iv. TELOPHASE
In this phase, chromatids reach at the opposite poles and two sets of chromosomes are formed. Nucleus membrane and nucleolus reappear. This completes the nuclear division.

Q. What is mitosis? Describe its discovery and phases.

Ans. **MITOSIS**

Definition:

"The type of cell division in which a cell divides into two daughter cells, each with same number of chromosomes as were present in the parent cell is called mitosis."

DISCOVERY:

In 1880s, German biologist, Walther Fleming observed that in a dividing cell, nucleus passes through a series of changes which he called mitosis.

OCURRENCE:

- Mitosis occurs only in eukaryotic cells.
- In multicellular organisms, the somatic cells undergo mitotic.

The process of mitosis is divided into two major phases.

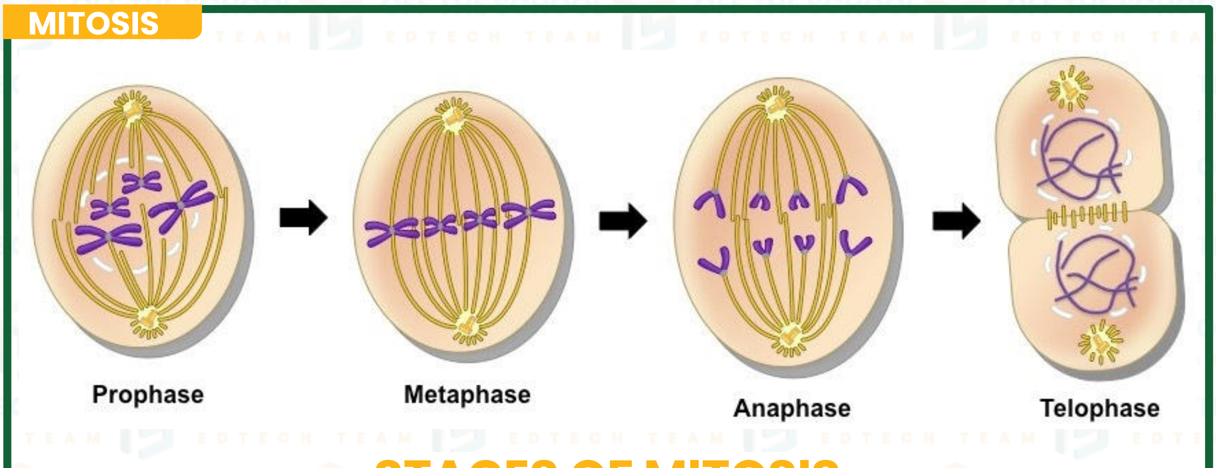
a. KARYOKINESIS:

The division of nucleus known as karyokinesis is divided into four phases.

i. Prophase.
ii. Metaphase.
iii. Anaphase.
iv. Telophase.

b. CYTOKINESIS:

The division of cytoplasm is known as cytokinesis.



STAGES OF MITOSIS

Q. Define Meiosis. Describe the stages of meiosis –I with suitable diagrams

Ans. **MEIOSIS:**

DEFINITION:

It is a type of cell division in which a parent cell divided into four daughter cells each having half number of chromosomes as contained in parent cell. Meiosis completes in two stages and occurs in germ cells.

EVENTS OF MITOSIS:

Meiosis consists of two successive divisions of a mother cell. The first division is a reduction division in which number of chromosomes reduces to half while second division is just like mitosis. Each meiotic division is further divided into four phases.

FIRST MEIOTIC DIVISION:

Following events takes place in first meiotic division.

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1. PROPHASE I:

The prophase I is the longest phase of meiosis and it's divided into the following Sub-stages.

a. LEPTOTENE:

In this stage, chromosomes are appeared to be thin, long thread like and longitudinally single.

b. ZYOGTENE:

During this stage, homologous chromosomes move towards each other and become ultimatelyassociated to form bivalents. This process of pairing of homologous chromosomes is called synapsis.

c. PACHYTENE:

During this stage, the paired chromosomes of he each bivalent get shortened and thickened. Homologous chromosomes get coiled around each other and each starts splitting longitudinally to form four chromatids called Tetrad.

d. DIPLOTENE:

During this stage, chromosomes uncoil and get separated. This separation is incomplete and paired chromosomes are in contact with each other at one or more points called Chiasmata. During uncoiling exchange of chromosomes parts occurs. This exchange is called Crossing over.

e. DIAKINESIS:

During this sub-stage, nuclear membrane and nucleus disappears. The separation of bivalents is completed by the process of terminalization, by which chiasmata move from centromere towards the ends of chromosomes like a zipper.

2. METAPHASE I:

During this stage, bivalent is arranged at equatorial plane, and each attach itself to the half spindle fibers.

3. ANAPHASE I:

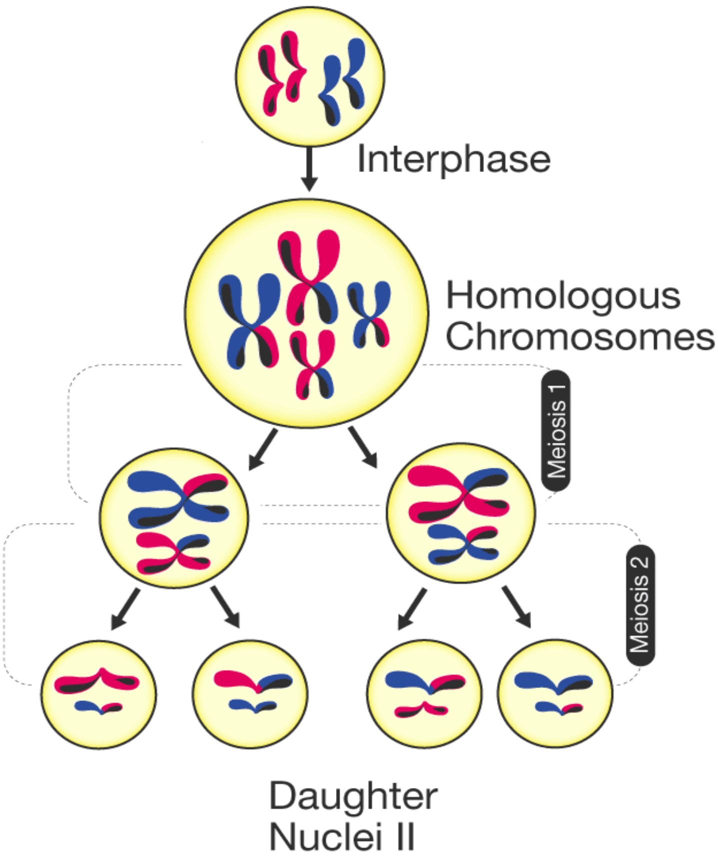
In the contraction, of spindle fibers result in movement of each chromosome of homologous pairs towards the opposite. Hence two sets of chromosomes are formed at opposite poles of the cell.

4. TELOPHASE I:

Now, nuclear membrane is formed around each of set of chromosomes and nucleolus reappears.

SOMATIC CELLS	GERM LINE CELLS
Somatic cells are those which form the body of organisms.	Germ line cells are those which give rise to gametes.
Somatic cells undergo mitosis.	Germ line cells undergo meiosis.

MEIOSIS



STAGES OF MEIOSIS

Q. Write the significance of Mitosis.

Ans. **SIGNIFICANCE OF MITOSIS:**

• Mitosis plays an important role in the life of an organisms. It is responsible for development and growth of organisms by increasing exact copies of cells.

• The production of new cells (Somatic cells), such as blood cells depend on mitosis. The healing of wounds, repair of wear and tear within organisms is also dependent upon the mitotic division.

Q. Define Apoptosis and Necrosis.

Ans. **APOPTOSIS:**

DEFINITION:

The type of cell which is programmed change which lead to sequence of physiological changes in cell by which cells commit suicide collectively called Apoptosis.

NECROSIS:

DEFINITION:

The type of cell death which is caused by external factors i.e., infection, toxin and tumor i.e., accidental cell death.

Q. Write the significance of meiosis.

Ans. **SIGNIFIANCE OF MEIOSIS:**

As a result of meiosis, gametes are formed, each of which possess haploid number of chromosomes, At the time of fertilization, chromosomes number becomes diploid. Thus, meiosis helps in maintaining the chromosomal number in specie generation after generation. During the process of meiosis, crossing over takes place, which results in new combination of genetical materials.

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Q. What is meiotic error (no question in notes)?
Ans. **MEIOTIC ERROR:**
During meiosis, two chromosomes of each homologous pair separate out and enters two separate cells, but sometimes a pair of homologous chromosomes fails to separate from one another, during meiosis I. This phenomenon is called **NON-DISJUNCTION**. Non-disjunction produces gametes with abnormal number of chromosomes i.e., either with less or extra chromosomes. If such abnormal gametes fuse with normal gametes, the **SCIENTIFIC REASONS**
Q.1. How number of chromosomes remains constant from generation to generation?
Ans. Meiosis maintain chromosomes number constant from generation to generation. It is due to the fact that meiosis reduces the diploid number of chromosomes to half i.e. haploid in the gametes and during fertilization the diploid number of the chromosomes is restored.
Q.2. Why inter-phase is called as phase of high metabolic activities?
Ans. Interphase is said to be high metabolic activity phase due to the fact cell grow in size, specific enzymes are synthesized and DNA base units are accumulated for DNA synthesis in this phase.
Q.3. Why inter-phase between meiosis- I and meiosis- II is short?
Ans. Interphase between meiosis- I and meiosis- II is relatively short because the DNA replication does not occur. Replication of DNA is not needed because the chromosomes already have two chromatids